



Qualification Pack

CNC Turning Programmer

QP Code: CUTM/CSC/Q0201

Version: 1.0

NSQF Level: 5

Centurion University of Technology and Management || Aluri Nagar, Paralakhemundi, R Sitapur
Uppalada || email:partha@cutm.ac.in



Qualification Pack

Contents

CUTM/CSC/Q0201: CNC Turning Programmer	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
CUTM/CSC/N0201: Interpretation of designs, drawings and process planning.	5
CUTM/CSC/N0202: Design CAM Programs, metrology, maintenance of machineries	10
CUTM/CSC/N0203: Operate CNC Turning machine	14
CUTM/CSC/N0204: Inspect, Finalize and deliver the parts	19
CUTM/VSQ/N7805: Ensure and Implement Safety standards at Work Place	24
DGT/VSQ/N0102: Employability Skills (60 Hours)	28
Assessment Guidelines and Weightage	35
<i>Assessment Guidelines</i>	35
<i>Assessment Weightage</i>	37
Acronyms	38
Glossary	39

Qualification Pack

CUTM/CSC/Q0201: CNC Turning Programmer

Brief Job Description

CNC Turning Programmer is responsible for understanding the design requirements, material, generate CAM design for manufacturing of various parts using CNC lathe, inspect the parts to finalize and deliver the finished product. Also maintain good condition of the machine with regular maintenance.

Personal Attributes

A CNC Turning Programmer should be precise, analytical, and technically skilled. They must possess strong interpretation abilities for technical drawings and CAM programming. Attention to detail, problem-solving, and machine-handling expertise are essential. Good time management, commitment to quality, adaptability to new technologies, and the ability to work independently or in teams are crucial for this role.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [CUTM/CSC/N0201: Interpretation of designs, drawings and process planning.](#)
2. [CUTM/CSC/N0202: Design CAM Programs, metrology, maintenance of machineries](#)
3. [CUTM/CSC/N0203: Operate CNC Turning machine](#)
4. [CUTM/CSC/N0204: Inspect, Finalize and deliver the parts](#)
5. [CUTM/VSQ/N7805: Ensure and Implement Safety standards at Work Place](#)
6. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Capital Goods
Sub-Sector	Advanced Machining
Occupation	Machining
Country	India
NSQF Level	5
Credits	19

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Aligned to NCO/ISCO/ISIC Code	NSQF/NCrF
Minimum Educational Qualification & Experience	Completed 3 year UG degree (B.E. or B. Tech degree in Mechanical, Production, Mechatronics) with NA of experience OR Completed 2nd year diploma after 12th (Mechanical or production) with NA of experience OR Completed 3 year diploma after 10th (Mechanical or production) with 1.5 years of experience relevant experience OR Previous relevant Qualification of NSQF Level (4) with 3 Years of experience relevant experience OR Previous relevant Qualification of NSQF Level (4.5) with 1.5 years of experience relevant experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	20 Years
Last Reviewed On	NA
Next Review Date	30/01/2027
NSQC Approval Date	31/01/2024
Version	1.0
Reference code on NQR	QG-05-CG-01975-2024-V1-CUTM
NQR Version	1.0

Qualification Pack

CUTM/CSC/N0201: Interpretation of designs, drawings and process planning.

Description

This NOS focuses on understanding engineering drawings and technical specifications. It involves analyzing part requirements, selecting appropriate machining processes, determining tool paths, and planning operations to ensure precision, efficiency, and compliance with design standards in CNC turning.

Scope

The scope covers the following :

- This NOS includes analyzing engineering drawings, selecting machining processes, determining tool paths, planning operations, optimizing cutting parameters, and ensuring precision manufacturing while adhering to quality standards and production

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Interpret technical drawings and specifications of the product.
- PC2.** Identify critical features of the product, surface finish and geometric requirements.
- PC3.** Analyze manufacturability with the available resources.
- PC4.** Identify and prepare for 3D mental representation of parts.
- PC5.** Identify the materials for turning and their availability.
- PC6.** Identify critical sequences (with a high possibility of damage or unsafe practice)
- PC7.** Prepare for production of parts according to specifications.
- PC8.** Imagine solutions with reference to the capacities of the workshop environment and the required work (size of batch, complexity)
- PC9.** Identify the features of work pieces and the appropriate machining and measuring processes.
- PC10.** Identify and prepare the best work holding methods.
- PC11.** Identify, prepare, and calibrate the correct measuring tools.
- PC12.** Identify and prepare the correct cutting tools.
- PC13.** Identify critical sections (with a high possibility of damage or unsafe practice) and consider alternatives.
- PC14.** choose innovative ways of using the environment to solve technical issues.
- PC15.** Check if the potential solutions will be reliable until the end of the process.
- PC16.** Weigh each solution and choose the best based on the various parameters: speed, safety, price
- PC17.** Plan the sequence of operations and machining strategy.
- PC18.** Create awareness actions for critical operations where no alternative is available.

Knowledge and Understanding (KU)

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The individual on the job needs to know and understand:

- KU1.** Common engineering drawing symbols, notations, and GD&T (Geometric Dimensioning and Tolerancing).
- KU2.** Interpretation of 2D and 3D CAD models for CNC turning operations.
- KU3.** Material properties and their impact on machining processes.
- KU4.** Selection and application of cutting tools, inserts, and tool holders.
- KU5.** CNC programming fundamentals, including G-codes, M-codes, and cycles.
- KU6.** Process planning for CNC turning, including work holding and machining sequences.
- KU7.** Calculation of cutting parameters such as feed rate, spindle speed, and depth of cut.
- KU8.** Tolerances and surface finish requirements for precision machining.
- KU9.** Machine capabilities, limitations, and setup procedures for CNC lathes.
- KU10.** Safety protocols and best practices in CNC machining and process planning.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Interpret engineering drawings and technical specifications accurately.
- GS2.** Analyze and plan machining sequences based on design requirements.
- GS3.** Use CAD/CAM software to visualize and modify part designs for CNC turning.
- GS4.** Calculate machining parameters such as spindle speed, feed rate, and depth of cut.
- GS5.** Identify suitable tools, inserts, and work-holding methods for efficient machining.
- GS6.** Communicate machining plans, tolerances, and process requirements with team members.
- GS7.** Troubleshoot design-related issues and suggest modifications for manufacturability.
- GS8.** Maintain accurate documentation of process plans, tool selections, and setup sheets.
- GS9.** Adapt to new CNC programming methods and emerging machining technologies.
- GS10.** Follow workplace safety standards and quality control procedures in process planning.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	40	-	30
PC1. Interpret technical drawings and specifications of the product.	2	3	-	2
PC2. Identify critical features of the product, surface finish and geometric requirements.	2	3	-	2
PC3. Analyze manufacturability with the available resources.	2	3	-	2
PC4. Identify and prepare for 3D mental representation of parts.	1	2	-	1
PC5. Identify the materials for turning and their availability.	1	3	-	1
PC6. Identify critical sequences (with a high possibility of damage or unsafe practice)	1	2	-	1
PC7. Prepare for production of parts according to specifications.	1	2	-	1
PC8. Imagine solutions with reference to the capacities of the workshop environment and the required work (size of batch, complexity)	1	2	-	1
PC9. Identify the features of work pieces and the appropriate machining and measuring processes.	1	2	-	1
PC10. Identify and prepare the best work holding methods.	1	2	-	1
PC11. Identify, prepare, and calibrate the correct measuring tools.	2	2	-	2
PC12. Identify and prepare the correct cutting tools.	2	2	-	2
PC13. Identify critical sections (with a high possibility of damage or unsafe practice) and consider alternatives.	2	2	-	2

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. choose innovative ways of using the environment to solve technical issues.	3	2	-	2
PC15. Check if the potential solutions will be reliable until the end of the process.	3	2	-	3
PC16. Weigh each solution and choose the best based on the various parameters: speed, safety, price	2	2	-	2
PC17. Plan the sequence of operations and machining strategy.	2	2	-	2
PC18. Create awareness actions for critical operations where no alternative is available.	1	2	-	2
NOS Total	30	40	-	30

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/CSC/N0201
NOS Name	Interpretation of designs, drawings and process planning.
Sector	Capital Goods
Sub-Sector	
Occupation	Machining
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	30/01/2027
NSQC Clearance Date	31/01/2024

Qualification Pack

CUTM/CSC/N0202: Design CAM Programs, metrology, maintenance of machineries

Description

This NOS focuses on creating efficient CAM programs, applying metrology techniques for precision measurement, and performing routine maintenance of CNC machines. It ensures accurate machining, minimizes errors, and enhances machine performance for optimal production efficiency.

Scope

The scope covers the following :

- This NOS includes developing CAM programs for CNC turning, applying metrology techniques for precision measurement, performing machine maintenance, troubleshooting issues, optimizing machining processes, and ensuring compliance with quality and safety standards for efficient manufacturing operations.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** select the best sequence for machining according to the production type and part specification.
- PC2.** use skill specific software and related hardware.
- PC3.** generate programs using CAD/CAM systems.
- PC4.** create CNC programs using the provided drawings and software.
- PC5.** transfer programs to machines.
- PC6.** create or edit programs directly on the control (without CAD/CAM)
- PC7.** set work pieces and tools.
- PC8.** select appropriate measuring- or gauging instruments.
- PC9.** calibrate measuring tools.
- PC10.** use selected tools to make measurements on all components on drawings.
- PC11.** understand properties, uses, and handling of ferrous and non-ferrous materials.
- PC12.** do preventative maintenance to avoid issues arising.
- PC13.** troubleshoot minor repairs in the machinery.
- PC14.** escalate to the appropriate team if there is a major problem.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Common principles of Computer-Aided Manufacturing (CAM) and its application in CNC turning.
- KU2.** Selection and configuration of CAM software for generating CNC programs.

Qualification Pack

- KU3.** Toolpath generation techniques for roughing, finishing, and threading operations.
- KU4.** Methods of workpiece setup, tool offset, and fixture selection in CNC turning.
- KU5.** Fundamentals of metrology, including the use of micrometers, calipers, dial indicators, and CMM (Coordinate Measuring Machine).
- KU6.** Interpretation of inspection reports, tolerance analysis, and quality control standards.
- KU7.** Preventive and corrective maintenance of CNC machines, including lubrication, alignment, and troubleshooting.
- KU8.** Optimization of cutting parameters to improve tool life and machining efficiency.
- KU9.** Troubleshooting common errors in CAM programming and CNC operations.
- KU10.** Safety standards and best practices in CNC machining, tool handling, and machine maintenance.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Apply logical thinking and problem-solving skills to design efficient CAM programs.
- GS2.** Interpret engineering drawings and CAD models for accurate tool path generation.
- GS3.** Use CAM software to create, modify, and simulate CNC turning programs.
- GS4.** Select appropriate cutting tools, speeds, and feeds based on material properties.
- GS5.** Utilize precision measuring instruments for dimensional accuracy and quality control.
- GS6.** Identify machine maintenance requirements and perform preventive maintenance tasks.
- GS7.** Troubleshoot errors in machining programs and adjust parameters accordingly.
- GS8.** Ensure compliance with safety protocols and industry standards during operations.
- GS9.** Document programming details, inspection reports, and maintenance logs.
- GS10.** Communicate effectively with team members, supervisors, and quality inspectors.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	40	60	-	-
PC1. select the best sequence for machining according to the production type and part specification.	3	4	-	-
PC2. use skill specific software and related hardware.	3	4	-	-
PC3. generate programs using CAD/CAM systems.	3	4	-	-
PC4. create CNC programs using the provided drawings and software.	3	4	-	-
PC5. transfer programs to machines.	2	4	-	-
PC6. create or edit programs directly on the control (without CAD/CAM)	3	4	-	-
PC7. set work pieces and tools.	3	4	-	-
PC8. select appropriate measuring- or gauging instruments.	3	4	-	-
PC9. calibrate measuring tools.	3	4	-	-
PC10. use selected tools to make measurements on all components on drawings.	3	4	-	-
PC11. understand properties, uses, and handling of ferrous and non-ferrous materials.	3	5	-	-
PC12. do preventative maintenance to avoid issues arising.	3	5	-	-
PC13. troubleshoot minor repairs in the machinery.	3	5	-	-
PC14. escalate to the appropriate team if there is a major problem.	2	5	-	-
NOS Total	40	60	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/CSC/N0202
NOS Name	Design CAM Programs, metrology, maintenance of machineries
Sector	Capital Goods
Sub-Sector	
Occupation	Machining
NSQF Level	5
Credits	4
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	30/01/2027
NSQC Clearance Date	31/01/2024

Qualification Pack

CUTM/CSC/N0203: Operate CNC Turning machine

Description

This NOS focuses on setting up, programming, and operating CNC turning machines. It includes selecting tools, loading workpieces, executing machining processes, monitoring operations, ensuring dimensional accuracy, troubleshooting issues, and following safety and quality standards for efficient production.

Scope

The scope covers the following :

- This NOS includes setting up CNC machines, selecting tools, loading workpieces, executing machining programs, monitoring operations, ensuring dimensional accuracy, troubleshooting issues, maintaining safety protocols, and optimizing processes to achieve high-quality production in compliance with industry standards.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Follow their selected process strategy.
- PC2.** Follow a given process- strategy (external CNC program)
- PC3.** Upload generated CNC programs to CNC lathes and perform test runs.
- PC4.** Mount and align selected tools.
- PC5.** Mount and align selected work holding devices.
- PC6.** Mount and align selected accessories (Tailstock, Parts catcher, etc.)
- PC7.** Avoid vibration in the machining sequences.
- PC8.** Apply burr-removal techniques on work pieces.
- PC9.** Optimize machining strategies.
- PC10.** Quickly react if anything goes wrong (making emergency stops)
- PC11.** obtain dimensions, geometries, surface roughness by interacting with CNC lathes.
- PC12.** Make the final parts conform to blueprints.
- PC13.** Report health, safety, and environmental issues to the appropriate personnel.
- PC14.** Report equipment failure to the appropriate personnel.
- PC15.** Note all the production related information on the respective register, log, work sheet.
- PC16.** Update all production information on the MIS system.
- PC17.** Store the CNC program against the batch number for future reference.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Fundamentals of CNC turning operations, including coordinate systems, tool paths, and machining principles.

Qualification Pack

- KU2.** Understanding of engineering drawings, GD&T (Geometric Dimensioning and Tolerancing), and symbols used in CNC programming.
- KU3.** Knowledge of different types of cutting tools, tool holders, inserts, and their selection based on workpiece material.
- KU4.** Awareness of cutting parameters such as speed, feed, depth of cut, and their impact on machining quality.
- KU5.** Knowledge of common CNC codes (G-codes, M-codes) and their functions in CNC turning programs.
- KU6.** Understanding CNC machine components, including spindle, turret, chuck, tailstock, and coolant system.
- KU7.** Ability to set up workpieces, tool offsets, and zero references to ensure accurate machining operations.
- KU8.** Knowledge of troubleshooting techniques for CNC machine errors, alarms, and tool wear issues.
- KU9.** Awareness of standard operating procedures (SOPs), safety guidelines, and workplace best practices in CNC machining.
- KU10.** Understanding quality control measures, inspection tools (calipers, micrometers, gauges), and documentation requirements for CNC-turned components.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret engineering drawings, work instructions, and CNC programs accurately.
- GS2.** coordinate with supervisors, engineers, and team members for smooth workflow.
- GS3.** keep documentation and report-writing skills for recording machining parameters, tool life, and quality checks.
- GS4.** make informed decisions regarding tool selection, machining parameters, and troubleshooting machine errors.
- GS5.** plan machining operations, managing tool inventory, and scheduling tasks efficiently.
- GS6.** Understand customer requirements, quality standards, and delivery expectations for precision components.
- GS7.** identify and resolve machining defects, tool wear issues, and CNC programming errors.
- GS8.** analyze machining data, optimize cutting parameters, and improve productivity.
- GS9.** Evaluate process efficiency, identify areas for improvement, and implementing corrective actions.
- GS10.** Work effectively in a manufacturing environment, coordinating with quality inspectors, maintenance teams, and production staff.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	40	60	-	-
PC1. Follow their selected process strategy.	2	3	-	-
PC2. Follow a given process- strategy (external CNC program)	3	3	-	-
PC3. Upload generated CNC programs to CNC lathes and perform test runs.	2	3	-	-
PC4. Mount and align selected tools.	3	3	-	-
PC5. Mount and align selected work holding devices.	3	3	-	-
PC6. Mount and align selected accessories (Tailstock, Parts catcher, etc.)	3	3	-	-
PC7. Avoid vibration in the machining sequences.	3	3	-	-
PC8. Apply burr-removal techniques on work pieces.	3	3	-	-
PC9. Optimize machining strategies.	2	4	-	-
PC10. Quickly react if anything goes wrong (making emergency stops)	2	4	-	-
PC11. obtain dimensions, geometries, surface roughness by interacting with CNC lathes.	2	4	-	-
PC12. Make the final parts conform to blueprints.	2	4	-	-
PC13. Report health, safety, and environmental issues to the appropriate personnel.	2	4	-	-
PC14. Report equipment failure to the appropriate personnel.	2	4	-	-
PC15. Note all the production related information on the respective register, log, work sheet.	2	4	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC16. Update all production information on the MIS system.	2	4	-	-
PC17. Store the CNC program against the batch number for future reference.	2	4	-	-
NOS Total	40	60	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/CSC/N0203
NOS Name	Operate CNC Turning machine
Sector	Capital Goods
Sub-Sector	
Occupation	Machining
NSQF Level	5
Credits	4
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	30/01/2027
NSQC Clearance Date	31/01/2024

Qualification Pack

CUTM/CSC/N0204: Inspect, Finalize and deliver the parts

Description

This NOS focuses on conducting precision inspections using metrology tools, verifying dimensions against design specifications, identifying defects, performing necessary finishing operations, documenting quality checks, and ensuring proper packaging and labeling for final delivery while meeting industry standards.

Scope

The scope covers the following :

- This NOS includes conducting precision inspections using metrology tools, verifying dimensions against design specifications, identifying defects, performing finishing operations, documenting quality checks, ensuring proper packaging, and delivering defect-free parts while adhering to industry quality and safety standards..

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** check visual inspection on completion of all the process as per drawing.
- PC2.** check the calibration status of various measuring instrument.
- PC3.** identify parameter wise measuring instruments for measurement.
- PC4.** inspect the machined product with various measurement tools like Vernier caliper and micrometer.
- PC5.** use coordinate Measuring Machine to inspect the profile measurements.
- PC6.** use height gauge for appropriate measurements.
- PC7.** check the surface finish as per requirements.
- PC8.** check the fitness of the product using go or no-go devices.
- PC9.** check visual inspection on the damage, burr, excess materials.
- PC10.** document all the measured data for various parameters.
- PC11.** identify the out of dimension parts.
- PC12.** check for rework possibilities and complete the rework.
- PC13.** reject the parts not able to rework.
- PC14.** Complete the documentation as per ISO Standards.
- PC15.** pack the parts in proper way to avoid any damages during transport.
- PC16.** complete the delivery documents.
- PC17.** deliver the parts, drawings and digital memory devices to the appropriate location and/or personnel as required by the organization
- PC18.** dismount tools, clamping, accessories.
- PC19.** Use Statistical Process Control tools to monitor the process.
- PC20.** set the environment to its initial state, ready for the next job.

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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** engineering drawings.
- KU2.** surface finish requirements, material properties, and post-machining treatments such as deburring and polishing.
- KU3.** quality control standards and statistical process control (SPC) techniques.
- KU4.** set up and use inspection tools such as height gauges, bore gauges, and profile projectors for dimensional verification.
- KU5.** common machining defects (e.g., tool marks, burrs, out-of-tolerance dimensions) and methods to rectify them before finalizing parts.
- KU6.** documentation and reporting, including preparing inspection reports, rejection reports, and maintaining quality records.
- KU7.** standard operating procedures (SOPs) for inspection, packaging, and delivery of finished parts.
- KU8.** customer specifications, order requirements, and delivery schedules to ensure timely dispatch of quality parts.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret engineering drawings and inspection reports accurately.
- GS2.** discuss inspection results, deviations, and corrective actions with supervisors and team members.
- GS3.** maintain quality records, rejection reports, and final delivery checklists.
- GS4.** decide whether a part meets quality standards or requires rework based on inspection data.
- GS5.** schedule inspections, maintain inspection tools, and coordinate delivery timelines.
- GS6.** Understand customer quality expectations and ensuring defect-free parts are delivered.
- GS7.** Identify machining defects, analyzing root causes, and suggesting corrective actions.
- GS8.** evaluate inspection data, detect trends in defects, and suggest process improvements.
- GS9.** Assess part quality against standards and taking proactive measures to prevent quality issues.
- GS10.** Work effectively with CNC operators, quality inspectors, and logistics teams to ensure smooth finalization and dispatch of parts.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	40	60	-	-
PC1. check visual inspection on completion of all the process as per drawing.	2	3	-	-
PC2. check the calibration status of various measuring instrument.	2	3	-	-
PC3. identify parameter wise measuring instruments for measurement.	2	3	-	-
PC4. inspect the machined product with various measurement tools like Vernier caliper and micrometer.	2	3	-	-
PC5. use coordinate Measuring Machine to inspect the profile measurements.	2	3	-	-
PC6. use height gauge for appropriate measurements.	2	3	-	-
PC7. check the surface finish as per requirements.	2	3	-	-
PC8. check the fitness of the product using go or no-go devices.	2	3	-	-
PC9. check visual inspection on the damage, burr, excess materials.	2	3	-	-
PC10. document all the measured data for various parameters.	2	3	-	-
PC11. identify the out of dimension parts.	2	3	-	-
PC12. check for rework possibilities and complete the rework.	2	3	-	-
PC13. reject the parts not able to rework.	2	3	-	-
PC14. Complete the documentation as per ISO Standards.	2	3	-	-
PC15. pack the parts in proper way to avoid any damages during transport.	2	3	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC16. complete the delivery documents.	2	3	-	-
PC17. deliver the parts, drawings and digital memory devices to the appropriate location and/or personnel as required by the organization	2	3	-	-
PC18. dismount tools, clamping, accessories.	2	3	-	-
PC19. Use Statistical Process Control tools to monitor the process.	3	3	-	-
PC20. set the environment to its initial state, ready for the next job.	1	3	-	-
NOS Total	40	60	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/CSC/N0204
NOS Name	Inspect, Finalize and deliver the parts
Sector	Capital Goods
Sub-Sector	
Occupation	Machining
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	30/01/2027
NSQC Clearance Date	31/01/2024

Qualification Pack

CUTM/VSQ/N7805: Ensure and Implement Safety standards at Work Place

Description

This NOS focuses on developing the ability to operate CNC machining centers safely and efficiently while adhering to health and safety standards. It includes creating CAM programs, selecting tools and cutting parameters, maintaining machinery, communicating effectively with teams and clients, and continuously upgrading skills to ensure excellence in manufacturing processes.

Scope

The scope covers the following :

- The scope encompasses CNC machine operation, safety compliance, program development, tool selection, maintenance, and effective collaboration with team members and customers. It promotes high professional standards through ongoing skill enhancement, problem-solving, and customer engagement while maintaining health and safety protocols throughout the manufacturing lifecycle.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** adhere to health and safety regulations in the workplace
- PC2.** proactively promote best practice in health and safety in the working environment
- PC3.** work independently on CNC machining centres
- PC4.** create manual and CAM programs for various types of machining
- PC5.** select suitable cutting parameters
- PC6.** select and set the most appropriate tools for the planned work
- PC7.** maintain all tools to ensure that they are in the best condition
- PC8.** communicate and collaborate effectively with colleagues, team members, and other professionals
- PC9.** engage with customers effectively, always prioritizing their needs
- PC10.** explain complex technical details to non-specialists
- PC11.** proactively engage in continuous professional development to promote excellence in the work and maintain expertise in current industrial practice analyze the manufacturing feasibility
- PC12.** successfully apply mathematical principles to complex industrial scenarios
- PC13.** demonstrate high levels of critical thinking

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Workplace health and safety regulations, and how to adhere to them.
- KU2.** Best practices in promoting health and safety proactively in CNC operations.

Qualification Pack

- KU3.** Fundamentals of CNC machining processes and the safe operation of CNC machines.
- KU4.** CAM software programming principles and manual programming techniques.
- KU5.** Selection of appropriate cutting parameters and tools for various materials and tasks.
- KU6.** Maintenance procedures for tools and machines to ensure optimal performance and safety.
- KU7.** Communication protocols and collaborative practices within a technical team.
- KU8.** Techniques for engaging with customers and understanding their technical needs.
- KU9.** Simplifying and explaining technical information to non-specialists.
- KU10.** Importance of continuous professional development and industrial best practices.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Independently operate CNC machines with precision and safety.
- GS2.** Develop CAM programs suited for diverse machining requirements.
- GS3.** Select and calibrate tools and cutting parameters for optimal machining.
- GS4.** Maintain tools and machines in excellent working condition.
- GS5.** Collaborate effectively with teammates and interdisciplinary professionals.
- GS6.** Provide clear, simple explanations of technical content to clients or stakeholders.
- GS7.** Prioritize customer requirements and ensure satisfaction in service delivery.
- GS8.** Apply mathematical reasoning to solve complex machining problems.
- GS9.** Analyze the feasibility of manufacturing processes and adapt accordingly.
- GS10.** Exhibit strong critical thinking and decision-making in industrial scenarios.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	40	-	30
PC1. adhere to health and safety regulations in the workplace	2	3	-	2
PC2. proactively promote best practice in health and safety in the working environment	2	3	-	2
PC3. work independently on CNC machining centres	2	3	-	2
PC4. create manual and CAM programs for various types of machining	2	3	-	2
PC5. select suitable cutting parameters	2	3	-	2
PC6. select and set the most appropriate tools for the planned work	2	3	-	2
PC7. maintain all tools to ensure that they are in the best condition	2	3	-	2
PC8. communicate and collaborate effectively with colleagues, team members, and other professionals	2	3	-	2
PC9. engage with customers effectively, always prioritizing their needs	2	2	-	2
PC10. explain complex technical details to non-specialists	2	2	-	3
PC11. proactively engage in continuous professional development to promote excellence in the work and maintain expertise in current industrial practice analyze the manufacturing feasibility	3	4	-	3
PC12. successfully apply mathematical principles to complex industrial scenarios	3	4	-	3
PC13. demonstrate high levels of critical thinking	4	4	-	3
NOS Total	30	40	-	30



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/VSQ/N7805
NOS Name	Ensure and Implement Safety standards at Work Place
Sector	Cross Sectoral
Sub-Sector	
Occupation	Health and Safety
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	30/01/2027
NSQC Clearance Date	31/01/2024



Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Assessment System Overview:

- Batches assigned to the assessment agencies for conducting the assessment on SIDH or email
- Assessment agencies send the assessment confirmation to TP looping AB/AA to accept in SIDH
- Assessment agency deploys the ToA certified Assessor for executing the assessment
- AB monitors the assessment process & records

2. Testing Environment:

- Check the Assessment location, date and time
- If the batch size is more than 30, then there should be 2 Assessors.
- Check that the allotted time to the candidates to complete Theory & Practical Assessment is correct.



Qualification Pack

3. Assessment Quality Assurance levels/Framework:

- Question bank is created by the Subject Matter Experts (SME) are verified by the AB
- Questions are mapped to the specified assessment criteria.
- Assessor must be ToA certified & trainer must be ToT Certified

4. Types of evidence or evidence-gathering protocol:

- Time-stamped & geotagged reporting of the assessor from assessment location.
- Centre photographs with signboards and scheme specific branding
- Video Call verification/online proctoring will be done during the assessment from AB.

5. Method of verification or validation:

- Surprise visit to the assessment location

6. Minimum Percentage Required

- 70% overall to pass.
- Minimum 70% in each NOS

7. Reassessment: Candidates failing the assessment will be given one opportunity for reassessment after additional training.

8. Assessor Details: ToA Certified Assessor with relevant experience.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Minimum Passing % at NOS Level: 70

(Please note: A Trainee must score the minimum percentage for each NOS separately as well as on the QP as a

Qualification Pack

whole.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CUTM/CSC/N0201.Interpretation of designs, drawings and process planning.	30	40	-	30	100	20
CUTM/CSC/N0202.Design CAM Programs, metrology, maintenance of machineries	40	60	-	-	100	20
CUTM/CSC/N0203.Operate CNC Turning machine	40	60	-	-	100	20
CUTM/CSC/N0204.Inspect, Finalize and deliver the parts	40	60	-	-	100	15
CUTM/VSQ/N7805.Ensure and Implement Safety standards at Work Place	30	40	-	30	100	15
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	200	290	-	60	550	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.